

Adora PINN — Step 7: Prototype inversion

As-built notes · Ivan Milic · 2026-09-20 · step 7 of pinn/plan.md

Step 7 is an end-to-end inversion of the observed testset (step 5): starting from the pretrained FAL-C field (warm start), optimise the field weights to fit the Stokes observations, regularised by hydrostatic equilibrium and the top-boundary pressure. Files [pinn/invert.py](#), [pinn/step7_inversion.ipynb](#); result saved to [/dat/milic/adora_pinn_develop/invert](#)

1. Machinery

`invert.run_inversion` runs optax Adam on the field weights with the total loss $w_{\text{fit}} \chi^2 + w_{\text{he}} \mathcal{L}_{\text{HE}} + w_{\text{bnd}} \mathcal{L}_{\text{bnd}}$. **Fresh random samples every step** — fit pixels (`m_fit`), HE collocation points (`k_he`) and boundary pixels (`m_bnd`) — so the fit is stochastic over the map and HE is enforced over the whole domain in expectation; a separate FIXED monitor set gives clean loss curves. The forward model is `rt_forward.synth_field` (field $\rightarrow$ EOS $\rightarrow$ piecewise-linear RT), differentiated end-to-end.

2. Continuable training & scheduler

`run_inversion` takes either a params pytree (fresh) or a **checkpoint** dict and returns `{params, state, key, step, hist, monitor}`; feeding it back continues training — the Adam moments, RNG, LR-schedule step count and accumulated history all carry over (the notebook has a re-runnable “continue” cell). The learning rate follows an **exponential-decay schedule** (`lr_lr_decay`^{step/lr_transition}, floored) — the standard, open-ended choice for PINNs. Per-Stokes fit weights are **hand-set** (`sigma_stokes`, default $(10^{-2}, 5 \cdot 10^{-3}, 5 \cdot 10^{-3}, 5 \cdot 10^{-3}) \times \langle I_c \rangle$ — Q, U, V a few $\times$ tighter than I , not orders apart).

3. Results (2000 steps, one H100)

Misfit RMS(model – obs)/ $\langle I_c \rangle$, initial $\rightarrow$ final: I 0.198 $\rightarrow$ 0.050, Q $1.2 \cdot 10^{-3} \rightarrow 1.2 \cdot 10^{-3}$, U $1.2 \cdot 10^{-3}$, V $9.3 \cdot 10^{-3} \rightarrow 8.3 \cdot 10^{-3}$. The notebook shows the loss/LR curves, obs-vs-inverted I, Q, U, V profiles, and the recovered continuum, Doppler v_z and longitudinal B_z maps. Stokes I (the thermodynamics + velocity) is recovered well; the polarisation is only partly fit.

4. Known limitation & next

The recovered maps show **stripes** and $V/Q/U$ recovery is limited — diagnosed separately in [adora_pinn.7_debug.pdf](#): the random Fourier modes show through wherever the data under-constrains the field, and this is best cured with a **spatial-smoothness regularizer** (plus moderate σ , denser sampling, and regularizing/freezing weak channels). That regularizer is the natural next addition to [losses.py](#). Files uncommitted.